PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY, SOLAPUR

B.Sc. (ECS)-II Sem - III Practical Examination, October 2026

VSC2: Based on DSC2-3 and DSC2-4

Date & Time: _____ Max Marks: 30

Instructions:

I) Solve any one from each section.

II) Each question carries 10 marks.

III) The Journal and Viva-Voce carry 5 marks each.

Section-I

Q1) Write a Python script to draw a random sample of size 40 from a normal population of size 400 with mean 35 and standard deviation 7 by using the stratified random sampling method. Calculate the sample mean and sample standard deviation.

Q2) Find Karl pearson's correlation coefficient between income and expenditure from the following data and interpret the result.

income	1000	2000	2500	2700	3000	3300	4100
expenditure	800	1300	1800	1900	2100	2340	2450

Section-II

Q1) Write a program to demonstrate any three itertools functions.

Q2) Write a program to demonstrate the use of *args and **kwargs arguments.